

Time: 1 Hrs.

Total Marks: 40

Note: Marks are indicated against each question.

Question No. 1

(8)

If P, Q and R are three atomic variables, obtain the Principal Disjunctive Normal form and Principal Conjunctive Normal form for the given formula without using TT.

$$(P \rightarrow (Q \wedge R)) \vee (\sim P \rightarrow (Q \vee R))$$

Question no. 2

(8)

Determine if the following inference pattern is Valid OR Invalid. Produce evidence (write the steps) to confirm its validity/invalidity.

$$\frac{(p \vee q) \rightarrow (s \wedge t) \quad [\sim((\sim s) \vee (\sim t))] \rightarrow [(\sim r) \vee q]}{(p \vee q) \rightarrow (r \rightarrow q)}$$

Question no. 3

(2+6)

What is a partition on a set? Let $A = \{1, 2, 3, 4, 5\}$ and a relation R defined on A is $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5)\}$. Obtain the cells of the partition.

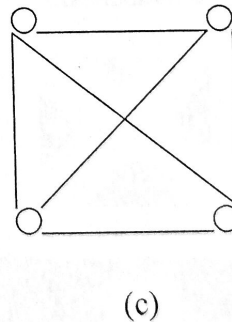
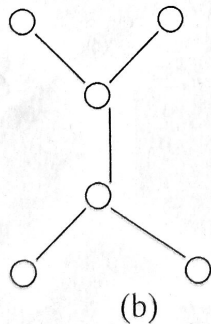
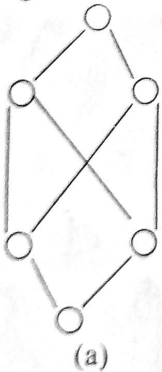
Question no. 4

(1+3+4)

What is a Hasse diagram? Draw the Hasse diagrams of the following sets under the partial ordering relation "divides" and indicate those which are totally ordered. Also, determine least member, greatest member, minimal member and maximal member in each case.

- (i) $\{2, 6, 24\}$
- (ii) $\{1, 2, 3, 6, 12\}$
- (iii) $\{3, 9, 27, 54\}$

Question no. 5 What is a lattice? Determine (show steps), which of the following graphs are lattice? (2+6)



SCHOOL OF COMPUTER & SYSTEMS SCIENCES, JNU
MCA I Semester Final Examination 2022-23
Discrete Mathematics

Time: 2 Hrs.

Total Marks: 60

Note: Attempt ALL Questions. Do Not Write Anything on the Question Paper.

Q. 1 If R is a relation "Less Than" from $A = \{1, 2, 3, 4\}$ to $B = \{1, 3, 5\}$ then what will be $R \circ R^{-1}$ (6)

Q. 2 R is a relation on set N defined by $\{(x, y) : x, y \in N, x + 2y = 21\}$. Determine the type of relation R ? (6)

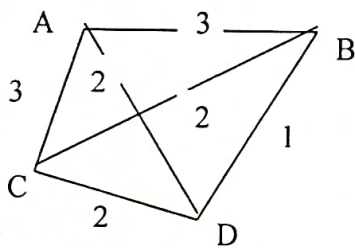
Q. 3 Define Boolean algebra. Prove that the power set of any set forms a Boolean algebra. (6)

Q. 4 There is a graph of 25 vertex (order) and 62 edges (size). The degree of every vertex is 3, 4, 5 or 6. There are 2 vertices of degree 4 and 11 vertices of degree 6. How many vertices of graph have degree 5? (6)

Q. 5 What is a spanning tree of a graph? How to obtain fundamental circuits, given a spanning tree? How many fundamental circuits exist in a graph? (6)

Q. 6 Explain the ringsum, fusion and deletion operations on graphs giving suitable examples. (6)

Q. 7 For the given graph, find the minimal spanning tree using Prim's algorithm (with steps). (6)



Q. 8 What do you understand by the planarity of a graph? Explain with example. State and prove the Euler formula to check the planarity of a graph. (6)

Q. 9 Define a Deterministic Finite Automaton (DFA)? Design a DFA for sentences in $\{a, b\}$ such that every a has b immediately to its right. (6)

Q. 10 Determine the language accepted by the following automaton. (6)

